24/01/2025
Morning

Aakash
Medical | IIT-JEE | Foundations

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Answers & Solutions

Time : 3 hrs. M.M. : 300

for

JEE (Main)-2025 Phase-1
[Computer Based Test (CBT) mode]

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (MPC) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three** Parts. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. Let $S_n = \frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\frac{1}{20}+\cdots$ upon n terms. If the sum of the first six terms of an A.P. with first term $-p$ and common difference p is $\sqrt{2026S_{2025}}$, then the absolute difference between 20th and 15th terms of the A.P. is

(1) 25 (2) 90

(3) 20 (4) 45

Answer (1)

Sol. $S_n = \frac{1}{1\times 2}+\frac{1}{2\times 3}+\cdots\frac{1}{n\times(n+1)}$

$=\left(\frac{1}{1}-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\cdots\left(\frac{1}{n}-\frac{1}{n+1}\right)$

$=1-\frac{1}{n+1}=\frac{n}{n+1}$

$\sqrt{S_{2025}\times 2026}=\sqrt{2026\times\frac{2025}{2026}}=45$

$S_6=\frac{6}{2}(2(-p)+5(p))=9p=45$

$\Rightarrow \quad p = 5$

$T_{20} - T_{15} = 5d = 5(p) = 25$

2. For a statistical data $x_1, x_2, \cdots, x_{10}$ of 10 values, a student obtained the mean as 5.5 and $\sum_{i=1}^{10} x_1^2 = 371$. He later found that he had noted two values in the data incorrectly as 4 and 5, instead of the correct values 6 and 8, respectively. The variance of the corrected data is

(1) 7 (2) 5

(3) 9 (4) 4

Answer (1)

Sol. $\sum x_i = (5.5)10 = 55$

$\sum x_i^2 = 371$

Corrected mean

$=\frac{\sum x_i'}{10}=\frac{55+6+8-(4+5)}{10}=6$

Now $\sum(x_i')^2 = 371+6^2+8^2-(4^2+5^2)$

= 471 – (41) = 430

New variance

$=\frac{\sum(x_i')^2}{10}-\left(\sum_{10} x_i'\right)^2=\frac{430}{10}-36=7$

3. Let in a ΔABC, the length of the side AC be 6, the vertex B be (1, 2, 3) and the vertices A, C lie on the line $\frac{x-6}{3}=\frac{y-7}{2}=\frac{z-7}{-2}$. Then the area (in sq. units) of ΔABC is:

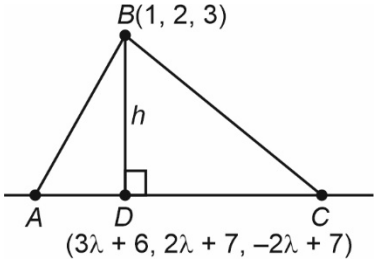
(1) 17 (2) 56

(3) 21 (4) 42

Answer (3)

Sol. Area = $\frac{1}{2}\times h(AC) = 3h$

Where h is minimum distance



$3(3\lambda + 5) + 2(2\lambda + 5) + (-2)(-2\lambda + 4) = 0$

$\Rightarrow \quad BD = 7 = h$

Area = 3(7) = 21



4. Let $f:\mathbb{R}-\{0\}\to\mathbb{R}$ be a function such that $f(x)-6f\left(\frac{1}{x}\right)=\frac{35}{3x}-\frac{5}{2}$. If the $\lim\limits_{x\to 0}\left(\frac{1}{\alpha x}+f(x)\right)=\beta$; $\alpha,\beta\in\mathbb{R}$, then $\alpha+2\beta$ is equal to

(1) 6 (2) 3

(3) 4 (4) 5

Answer (3)

Sol. $f(x)-6f\left(\frac{1}{x}\right)=\frac{35}{3x}-\frac{5}{2}$ ...(1)

$$6\left(f\left(\frac{1}{x}\right)-6f(x)=\frac{35x}{3}-\frac{5}{2}\right)$$

$$6f\left(\frac{1}{x}\right)-36f(x)=\frac{210x}{3}-\frac{30}{2} \quad ...(2)$$

(1) + (2)

$$-35f(x)=\frac{35}{3}\left[\frac{1}{x}+6x\right]-\frac{5}{2}(1+6)$$

$$-f(x)=\frac{1}{3}\left(\frac{1}{x}+6x\right)-\frac{1}{2}$$

$$f(x)=-\frac{1}{3x}-2x+\frac{1}{2}$$

$$\lim_{x\to 0}\left[\frac{1}{\alpha x}-\frac{1}{3x}-2x+\frac{1}{2}\right]=\beta$$

$$\Rightarrow \quad \alpha=3 \qquad \beta=\frac{1}{2}$$

$$\alpha+2\beta=3+2\times\frac{1}{2}=4$$

5. Let the line passing through the points (–1, 2, 1) and parallel to the line $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z}{4}$ intersect the line $\frac{x+2}{3}=\frac{y-3}{2}=\frac{z-4}{1}$ at the point P. Then the distance of P from the point $Q(4,-5,1)$ is

(1) 10 (2) $5\sqrt{5}$

(3) 5 (4) $5\sqrt{6}$

Answer (2)

Sol. Line passing through (–1, 2, 1) & parallel to $\frac{x-1}{2}=\frac{y+1}{3}=\frac{3}{4}$

$$L:\frac{x+1}{2}=\frac{y-2}{3}=\frac{z-1}{4}=\lambda$$

$$L_1:\frac{x+2}{3}=\frac{y-3}{2}=\frac{z-4}{1}=\mu$$

If lines are intersecting

$$\left.\begin{aligned}2\lambda-1&=3\mu-2\\3\lambda+2&=2\mu+3\end{aligned}\right\}\lambda=1 \quad \mu=1$$

$4\lambda + 1 = \mu + 4$

$\therefore$ Point $P(1, 5, 5)$ $Q(4, -5, 1)$

$$PQ=\sqrt{(1-4)^2+(5+5)^2+(5-1)^2}$$

$$=\sqrt{9+100+16}=\sqrt{125}=5\sqrt{5}$$

6. $\lim\limits_{x\to 0}\operatorname{cosec}x\left(\sqrt{2\cos^2 x+3\cos x}-\sqrt{\cos^2 x+\sin x+4}\right)$ is

(1) $\frac{1}{\sqrt{15}}$ (2) $-\frac{1}{2\sqrt{15}}$

(3) $\frac{1}{2\sqrt{15}}$ (4) 0

Answer (2)

Sol. $\lim\limits_{x\to 0}\operatorname{cosec}\,x\left(\sqrt{2\cos^2 x+3\cos x}-\sqrt{\cos^2 x+\sin x+4}\right)$

$$\lim_{x\to 0}\frac{\left(\sqrt{2\cos^2 x+3\cos x}-\sqrt{\cos^2 x+\sin x+4}\right)\left(\sqrt{2\cos^2 x+3\cos x}+\sqrt{\cos^2 x+\sin x+4}\right)}{\sin x\left(\sqrt{2\cos^2 x+3\cos x}+\sqrt{\cos^2 x+\sin x+4}\right)}$$

$$=\frac{1}{2\sqrt{5}}\lim_{x\to 0}\left[\frac{\cos^2 x+3\cos x-\sin x-4}{\sin x}\right]$$

Apply L' Hopital's rule

$$=\frac{1}{2\sqrt{5}}\lim_{x\to 0}\left[\frac{-\sin 2x-3\sin x-\cos x}{\cos x}\right]$$

$$=-\frac{1}{2\sqrt{5}}$$



7. Let $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}, \vec{b}=3\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}$ be three vectors such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$. If the vector $\vec{c}$ is perpendicular to $\vec{b}$ and $\vec{a}\cdot\vec{c}=5$, then $|\vec{c}|$ is equal to

(1) 18 (2) $\frac{1}{3\sqrt{2}}$

(3) $\sqrt{\frac{11}{6}}$ (4) 16

Answer (3)

Sol. $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$

$\vec{b}=3\hat{i}+\hat{j}-\hat{k}$

and $\vec{c} = k\left(\vec{a}+\lambda\vec{b}\right)$ (for some scalar λ)

$\therefore \quad \vec{c}=k\left((1+3\lambda)\hat{i}+(2+\lambda)\hat{j}+(3-\lambda)\hat{k}\right)$

$\because \ \vec{b}\cdot\vec{c}=0$

$3(1+3\lambda)+(2+\lambda)-(3-\lambda)=0$

$\therefore \quad \lambda=\frac{-2}{11}$

$\therefore \quad \vec{c}=k\left(\frac{5}{11}\hat{i}+\frac{20}{11}\hat{j}+\frac{35}{11}\hat{k}\right)$

$\because \quad \vec{c}\cdot\vec{a}=5 \Rightarrow k=\frac{11}{30}$

$\therefore \quad \vec{c}=\frac{1}{30}\left(5\hat{i}+20\hat{j}+35\hat{k}\right)$

$=\frac{1}{6}\left(\hat{i}+4\hat{j}+7\hat{k}\right)$

$\therefore \quad |\vec{c}|=\sqrt{\frac{11}{6}}$

8. Consider the region

$R=\left\{(x,y): x\le y\le 9-\frac{11}{3}x^2, x\ge 0\right\}$.

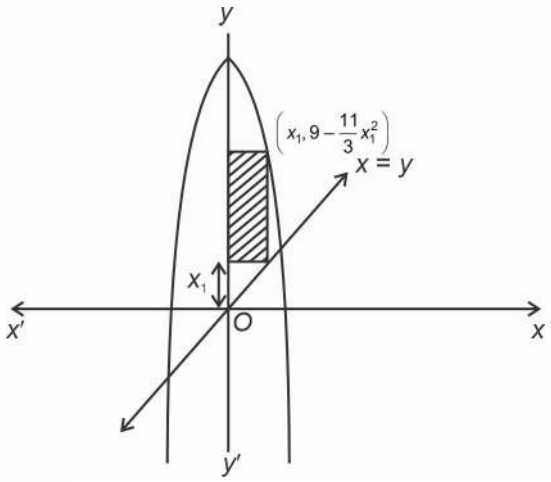
The area, of the largest rectangle of sides parallel to the coordinate axes and inscribed in R, is:

(1) $\frac{625}{111}$ (2) $\frac{821}{123}$

(3) $\frac{567}{121}$ (4) $\frac{730}{119}$

Answer (3)

Sol. $R=\left\{(x,y); x\le y\le 9-\frac{11}{3}x^2, x\ge 0\right\}$



Area of rectangle $(A) = \left(9-\frac{11}{3}x_1^2-x_1\right)x_1$

$A = 9x_1 - x_1^2 - \frac{11}{3}x_1^3$

$\frac{dA}{dx_1} = 9-2x_1-11x_1^2$

For maximum area = $x_1 = \frac{9}{11}$

$\therefore$ Required maximum area = $\frac{81}{11}-\frac{81}{121}-\frac{243}{121}$

$= \frac{567}{121}$ sq. units

9. The area of the region $\{(x, y) : x^2 + 4x + 2 \le y \le |x + 2|\}$ is equal to

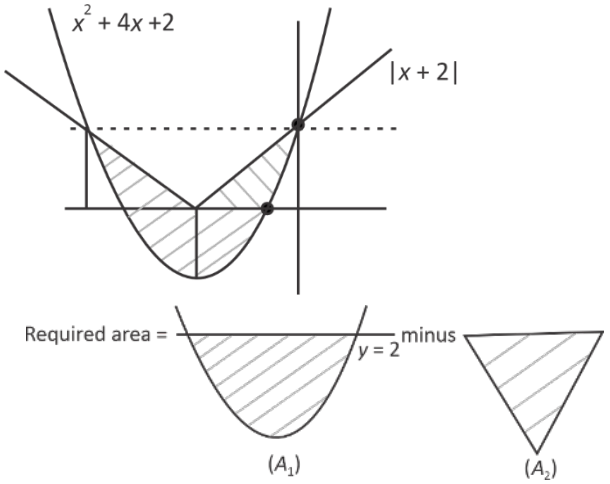
(1) 5 (2) $\frac{24}{5}$

(3) $\frac{20}{3}$ (4) 7

Answer (3)

Sol. $x^2 + 4x + 2 \le y \le |x + 2|$





$$A_1 = \int_{-4}^{0} [2 - (x^2 + 4x + 2)]dx - \frac{1}{2} \times 4 \times 2$$

$$= \left(\frac{-x^3}{3} - 2x^2 \right)\Bigg|_{-4}^{0} - 4$$

$$= 0 - \left(\frac{64}{3} - 32 \right) - 4$$

$$= 32 - \frac{64}{3} - 4 = \frac{20}{3}$$

10. *A* and *B* alternately throw a pair of dice. *A* wins if he throws a sum of 5 before *B* throws a sum of 8, and *B* wins if he throws a sum of 8 before *A* throws a sum of 5. The probability, that *A* wins if *A* makes the first throw, is

(1) $\frac{8}{19}$ (2) $\frac{9}{19}$

(3) $\frac{9}{17}$ (4) $\frac{8}{17}$

Answer (2)

Sol. For sum '5' $\rightarrow$ (1, 4), (2, 3), (3, 2)

$$(4, 1) \Rightarrow P(A) = \frac{4}{36}$$

For sum '8' $\rightarrow$ (2, 6), (3, 5), (4, 4)

$$(5, 3),\ (6, 2) \Rightarrow P(B) = \frac{5}{36}$$

$$P(\bar{A}) = \frac{32}{36}, P(\bar{B}) = \frac{31}{36}$$

$$P(A \text{ wins}) = P(A) + P(\bar{A})P(\bar{B})P(A) +$$

$$+P(\bar{A})P(\bar{B})P(\bar{A})P(\bar{B})P(A) + \ldots$$

$$= \frac{P(A)}{1 - P(\bar{A})P(\bar{B})} = \frac{9}{19}$$

11. Let $y = y(x)$ be the solution of the differential equation $\left(xy - 5x^2\sqrt{1+x^2}\right)dx + (1+x^2)dy = 0$, $y(0) = 0$. Then $y\left(\sqrt{3}\right)$ is equal to

(1) $\sqrt{\frac{14}{3}}$ (2) $\frac{5\sqrt{3}}{2}$

(3) $2\sqrt{2}$ (4) $\sqrt{\frac{15}{2}}$

Answer (2)

Sol. $\frac{dy}{dx} + \frac{xy}{1+x^2} = \frac{5x^2}{\sqrt{1+x^2}}$

I.F. $= e^{\int \frac{x}{1+x^2}dx} = \sqrt{1+x^2}$

$$y \cdot \sqrt{1+x^2} = \int \frac{5x^2}{\sqrt{1+x^2}} \cdot \sqrt{1+x^2}\,dx$$

$$y\sqrt{1+x^2} = \frac{5x^3}{3} + c$$

$y(0) = 0 \Rightarrow c = 0$

$$y\sqrt{1+x^2} = \frac{5x^3}{3}$$

$$y\left(\sqrt{3}\right) = \frac{5\sqrt{3}}{2}$$



12. If $I(m, n)=\int_0^1 x^{m-1}(1-x)^{n-1}dx, m, n>0$, then $I(9, 14) + I(10, 13)$ is

(1) $I(9, 1)$ (2) $I(1, 13)$

(3) $I(9, 13)$ (4) $I(19, 27)$

Answer (3)

Sol. Beta function

$$\beta(P, q)=\int_0^1 x^{P-1}(1-x)^{q-1}dx$$

$$=\frac{(P-1)!(q-1)!}{(P+q-1)!}, P, q \in I$$

$$I(9,14)=\frac{8!13!}{22!}$$

$$I(10,13)=\frac{9!12!}{22!}$$

$$I(9,14)+I(10,13)=\frac{8!12!}{21!}$$

$$=\frac{(9-1)!(13-1)!}{(9+13-1)!}=I(9,13)$$

13. The product of all the rational roots of the equation $(x^2 - 9x + 11)^2 - (x - 4)(x - 5) = 3$, is equal to

(1) 7 (2) 14

(3) 21 (4) 28

Answer (2)

Sol. $(x^2-9x+11)^2-(x-4)(x-5)=3$

$(x^2-9x+11)^2-(x^2-9x+20)=3$

Let $x^2-9x+11=t$

$t^2-(t+9)=3$

$\Rightarrow t^2-t-12=0$

$\Rightarrow t^2-4t+3t-12=0$

$\Rightarrow t(t-4)+3(t-4)=0$

$\Rightarrow t=4$ or -3

$x^2-9x+11=4$

$x^2-9x+7=0$

Here, we will get irrational roots

$x^2-9x+11=-3$

$x^2-9x+14=0$

$x^2-7x-2x+14=0$

$\Rightarrow x=7,2$

$\Rightarrow$ Product of all rational roots = 14

14. Let circle C be the image of $x^2 + y^2 - 2x + 4y - 4 = 0$ in the line $2x - 3y + 5 = 0$ and A be the point on C such that OA is parallel to x-axis and A lies on the right-hand side of the centre O of C. If $B(\alpha, \beta)$, with $\beta < 4$, lies on C such that the length of the arc AB is $(1/6)^{th}$ of the perimeter of C, then $\beta-\sqrt{3}\alpha$ is equal to

(1) 4 (2) 3

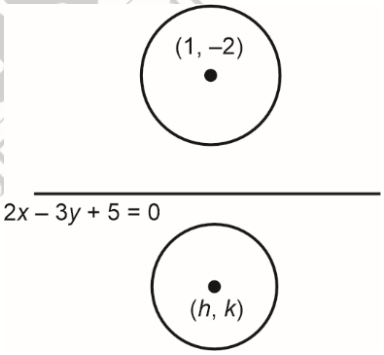
(3) $4-\sqrt{3}$ (4) $3+\sqrt{3}$

Answer (1)

Sol. $x^2+y^2-2x+4y-4=0$

Centre $\equiv(1, -2)$

Radius = $\sqrt{1+4+4}$ = 3



$$\frac{h-1}{2}=\frac{k+2}{-3}=\frac{-2(2+6+5)}{2^2+(-3)^2}$$

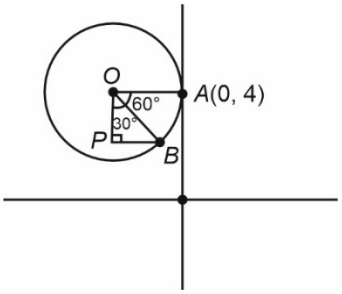
$$\Rightarrow \frac{h-1}{2}=\frac{k+2}{-3}=-\frac{26}{13}$$

$\Rightarrow h = -3, k = 4$

Centre of circle C (–3, 4)

$C : (x + 3)^2 + (y - 4)^2 = (3)^2$





Co-ordinate of $A(0, 4)$

As OA is parallel to x-axis length of are

$AB = \frac{1}{6}$ × perimeter

$= \frac{1}{6} \times 2\pi \times 3$

$\theta = \frac{\pi}{180} \times 3 = \pi$

$\Rightarrow \quad \theta = 60°$

In ΔOPB

$\sin 30° = \frac{PB}{OB}$

$\frac{PB}{3} = \frac{1}{2}$

$\Rightarrow \quad PB = \frac{-3}{2}$

$\cos 30° = \frac{OP}{3}$

$\Rightarrow \quad OP = \frac{3\sqrt{3}}{2}$

$B\left(\frac{-3}{2}, 4 - \frac{3\sqrt{3}}{2}\right)$

$\beta - \sqrt{3}\alpha = 4 - \frac{3\sqrt{3}}{2} + \frac{3\sqrt{3}}{2} = 4$

15. If α and β are the roots of the equation $2z^2 - 3z - 2i = 0$, where $i = \sqrt{-1}$, then $16 \cdot \text{Re}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right) . \text{Im}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right)$ is equal to

(1) 441 (2) 398

(3) 312 (4) 409

Answer (1)

Sol. $2z^2 - 3z - 2i = 0 \qquad \ldots(i)$

$2\left(z - \frac{i}{z}\right) = 3$

As α, β are roots of (i)

$\alpha - \frac{i}{\alpha} = \frac{3}{2}$

$\Rightarrow \alpha^2 - \frac{1}{\alpha^2} - 2i = \frac{9}{4}$

$\Rightarrow \alpha^2 - \frac{1}{\alpha^2} = \frac{9}{4} + 2i$

Squaring both sides

$\Rightarrow \alpha^4 + \frac{1}{\alpha^4} - 2 = \frac{81}{16} - 4 + 9i$

$\Rightarrow \alpha^4 + \frac{1}{\alpha^4} = \frac{49}{16} + 9i$

Similarly, $\beta^4 + \frac{1}{\beta^4} = \frac{49}{16} + 9i$

$$\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}$$

$$= \frac{\alpha^{15}\left(\alpha^4 + \frac{1}{\alpha^4}\right) + \beta^{15}\left(\beta^4 + \frac{1}{\beta^4}\right)}{\alpha^{15} + \beta^{15}}$$

$= \frac{49}{16} + 9i$

$\text{Re}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right) = \frac{49}{16}$

$\text{Im}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right) = 9$

$\Rightarrow 16\text{Re}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right) \cdot \text{Im}\left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}}\right)$

$= 16 \times \frac{49}{16} \times 9$

$= 441$



16. Let $f(x)=\frac{2^{x+2}+16}{2^{2x+1}+2^{x+4}+32}$. Then the value of $8\left(f\left(\frac{1}{15}\right)+f\left(\frac{2}{15}\right)+\ldots+f\left(\frac{59}{15}\right)\right)$ is equal to

(1) 118 (2) 92

(3) 102 (4) 108

Answer (1)

Sol. $f(x)=\frac{2^{x+2}+16}{2^{2x+1}+2^{x+4}+32}=\frac{4(2^x+4)}{2(2^{2x})+16(2^x)+32}$

$=\frac{2(2^x+4)}{\left(2^x+4\right)^2}=\frac{2}{2^x+4}$

Now, $f(x)+f(4-x)=\frac{2}{2^x+4}+\frac{2}{2^{4-x}+4}$

$=\frac{2}{2^x+4}+\frac{2\times 2^x}{16+4\times 2^x}=\frac{2}{2^x+4}+\frac{2x}{2\left(4+2^x\right)}$

$=\frac{1}{2}$

So, $f\left(\frac{1}{15}\right)+f\left(\frac{59}{15}\right)=\frac{1}{2}$

$f\left(\frac{2}{15}\right)+f\left(\frac{58}{15}\right)=\frac{1}{2}$

and $f\left(\frac{30}{15}\right)=\frac{2}{4+4}=\frac{1}{2}$

$\therefore 8\left(f\left(\frac{1}{15}\right)+f\left(\frac{2}{15}\right)+\ldots f\left(\frac{59}{15}\right)\right)$

$=8\left(\frac{29}{2}+\frac{1}{4}\right)=118$

17. If the system of equations

$2x-y+z=4$

$5x+\lambda y+3z=12$

$100x-47y+\mu z=212$

has infinitely many solutions, then $\mu-2\lambda$ is equal to

(1) 59 (2) 55

(3) 56 (4) 57

Answer (4)

Sol. $2x-y+z=4$

$5x+\lambda y+3z=12$

$100x-47y+\mu z=212$

$\Delta x=\Delta y=\Delta z=0$

$\Delta z=\begin{vmatrix}2 & -1 & 4\\ 5 & \lambda & 12\\ 100 & -47 & 212\end{vmatrix}=0$

$\Rightarrow 2(212\lambda+564)+1(1060-1200)$

$+4(-235-100\lambda)=0$

$\Rightarrow 424\lambda+1128-140-940-400\lambda=0$

$\Rightarrow \lambda=-2$

$\Delta y=\begin{vmatrix}2 & -1 & 1\\ 5 & -2 & 3\\ 100 & -47 & \mu\end{vmatrix}=0$

$\Rightarrow 2(-2\mu+141)+1(5\mu-300)+1(-235+200)=0$

$\Rightarrow \mu=53$

$\mu-2\lambda=53-2(-2)=57$

18. For some $n\neq 10$, let the coefficients of the 5th, 6th and 7th terms in the binomial expansion of $(1+x)^{n+4}$ be in A.P. Then the largest coefficient in the expansion of $(1+x)^{n+4}$ is:

(1) 10 (2) 70

(3) 35 (4) 20

Answer (3)

Sol. Binomial coefficient of $T_{r+1}={}^{n+4}C_r$

${}^{n+4}C_4$, ${}^{n+4}C_5$, ${}^{n+4}C_6$ in A.P.

$\Rightarrow \frac{(n+4)!}{4!\,n!}+\frac{(n+4)!}{6!\,(n-2)!}=2\frac{(n+4)!}{(n-1)!\,5!}$

$\Rightarrow n=3$

$\Rightarrow$ Greatest binomial coefficient in the expansion of $(1+x)^{n+4}$

$=(1+x)^7$

$\Rightarrow {}^7C_3={}^7C_4\Rightarrow \frac{7!}{3!\,4!}=\frac{7\times 6\times 5}{3\times 2}=35$



19. Let the product of the focal distances of the point $\left(\sqrt{3}, \frac{1}{2}\right)$ on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, (a>b)$, be $\frac{7}{4}$. Then the absolute difference of the eccentricities of two such ellipses is

(1) $\frac{3-2\sqrt{2}}{3\sqrt{2}}$ (2) $\frac{1-2\sqrt{2}}{\sqrt{3}}$

(3) $\frac{1-\sqrt{3}}{\sqrt{2}}$ (4) $\frac{3-2\sqrt{2}}{2\sqrt{3}}$

Answer (4)

Sol. Given $\left(a+e\sqrt{3}\right)\left(a-e\sqrt{3}\right)=\frac{7}{4}$ ...(i)

Also, $\frac{3}{a^2}+\frac{1}{4b^2}=1$ ...(ii)

And $b^2 = a^2(1-e^2)$ ...(iii)

From (i), (ii) and (iii)

$12e^4 - 17e^2 + 6 = 0$

$\Rightarrow (3e^2-2)(4e^2-3)=0 \Rightarrow e=\sqrt{\frac{2}{3}}$ or $\sqrt{\frac{3}{4}}$

20. Let the lines $3x-4y-\alpha=0$, $8x-11y-33=0$, and $2x-3y+\lambda=0$ be concurrent. If the image of the point (1, 2) in the line $2x-3y+\lambda=0$ is $\left(\frac{57}{13}, \frac{-40}{13}\right)$, then $|\alpha\lambda|$ is equal to

(1) 91 (2) 84

(3) 113 (4) 101

Answer (1)

Sol. $\begin{vmatrix} 3 & -4 & -\alpha \\ 8 & -11 & -33 \\ 2 & -3 & \lambda \end{vmatrix} = 0$

$\Rightarrow 2\alpha - \lambda = 33$ ...(i)

$\frac{x-1}{2}=\frac{y-2}{-3}=\frac{-2.|2-6+\lambda|}{4+9}=\frac{-2\lambda+8}{13}$

Image: $x=\frac{29-4\lambda}{13}=\frac{57}{13}$ (given)

$\Rightarrow \lambda = -7$

From (i) $\alpha = 13$

$|\alpha.\lambda| = 91$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. The number of 3-digit numbers, that are divisible by 2 and 3, but not divisible by 4 and 9, is

Answer (125)

Sol. Number of 3 digit number = 900

Number of 3 digit number divisible by 2 and 3

$= \frac{900}{6} = 150$

Number of 3 digit number divisible by 4 and 9

$= \frac{900}{36} = 25$

Number of 3 digit number divisible by 2 and 3 but not divisible by 4 and 9

$= 150 - 25 = 125$

22. Let f be a differentiable function such that $2(x+2)^2 f(x) - 3(x+2)^2 = 10\int_0^x (t+2)f(t)dt$, $x \geq 0$. Then $f(2)$ is equal to _____.

Answer (19)

Sol. $2(x+2)^2 f(x) - 3(x+2)^2 = 10\int_0^x (t+2)f(t)dt$

Differentiating both side

$4(x+2)f(x) + 2f'(x)(x+2)^2 - 6(x+2) = 10(x+2)f(x)$

$= (x+2)\frac{dy}{dx} - 3y = 3$

$\frac{1}{3}\int\frac{dy}{y+1}=\int\frac{dx}{x+2}$

$\ln|y+1| = 3\ln|x+2| + \ln c$

$y + 1 = (x+2)^3 c$

$\because y(0)=\frac{3}{2}$

$\Rightarrow \frac{5}{16}=c$

$\therefore \quad y=\frac{5}{16}(x+2)^3-1$

$y(2)=\frac{5}{16}\times 64 - 1 = 19$



23. Let A be a 3 × 3 matrix such that $X^T AX = O$ for all nonzero 3 × 1 matrices $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. If $A\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -5 \end{bmatrix}$, $A\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -8 \end{bmatrix}$, and $\det(\text{adj}(2(A + I))) = 2^{\alpha}3^{\beta}5^{\gamma}$. $\alpha, \beta, \gamma \in N$, then $\alpha^2 + \beta^2 + \gamma^2$ is_____.

Answer (44)

Sol. $X^T AX = O$

$$(xyz)\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$(xyz)\begin{bmatrix} a_1x + a_2y + a_3z \\ b_1x + b_2y + b_3z \\ c_1x + c_2y + c_3z \end{bmatrix} = 0$$

$x(a_1x + a_2y + a_3z) + y(b_1x + b_2y + b_3z)$
$+ z(c_1x + c_2y + c_3z)$

$a_1 = 0, b_2 = 0, c_3 = 0$

$a_2 + b_1 = 0, a_3 + c_1 = 0, b_3 + c_2 = 0$

A = skew symmetric matrix

$$A = \begin{bmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{bmatrix}; A\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -5 \end{bmatrix}$$

$x + y = 1$

$-x + z = 4$

$y + z = 5$

$$\begin{bmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -8 \end{bmatrix}$$

$2x + y = 0$ $\quad x = -1$

$-x + z = 4$ $\quad y = 2$

$-y - 2z = -8$ $\quad z = 3$

$$A = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$$

$$2(A + I) = \begin{bmatrix} 2 & -2 & 4 \\ 2 & 2 & 6 \\ -2 & -6 & 2 \end{bmatrix}$$

$2(A + I) = 120$

$\Rightarrow$ $\det(\text{adj}(2A + I))$
$= 120^2 = 2^6 . 3^2 . 5^2$

$\therefore$ $\alpha = 6, \beta = 2, \gamma = 2$

Hence $\alpha^2 + \beta^2 + \gamma^2 = 6^2 + 2^2 + 2^2 = 44$

24. Let $S = \{p_1, p_2 \ldots, p_{10}\}$ be the set of first ten prime numbers. Let $A = S \cup P$, where P is the set of all possible products of distinct elements of S. Then the number of all ordered pairs (x, y), $x \in S$, $y \in A$, such that x divides y, is _____.

Answer (5120)

Sol. Let $\frac{y}{x} = \lambda$

$y = \lambda x$

$= 10 \times \left({}^9C_0 + {}^9C_1 + {}^9C_2 + {}^9C_3 + \ldots + {}^9C_9\right)$

$= 10 \times 2^9$

$= 10 \times 512 = 5120$

25. If for some α, β; $\alpha \le \beta$, $\alpha + \beta = 8$ and $\sec^2(\tan^{-1}\alpha) + \text{cosec}^2(\cot^{-1}\beta) = 36$, then $\alpha^2 + \beta$ is _____.

Answer (14)

Sol. Let $\tan^{-1}\alpha = A \Rightarrow \tan A = \alpha$

$\cot^{-1}\beta = B \Rightarrow \cot B = \beta$

$\sec^2 A + \text{cosec}^2 B = 36$

$\Rightarrow$ $1 + \tan^2 A + 1 + \cot^2 B = 36$

$\Rightarrow$ $\alpha^2 + \beta^2 = 34$

Also $\alpha + \beta = 8$ (Given)

$\therefore$ $(\alpha + \beta)^2 = 34 + 2\alpha\beta = 64$

$\Rightarrow$ $\alpha\beta = 15$

$\Rightarrow$ α, β are roots of equation

$x^2 - 8x + 15 = 0$

$\Rightarrow$ $(x - 3)(x - 5) = 0$

$\Rightarrow$ $x = 3, 5$

$\therefore$ $\alpha = 3, \beta = 5$ $\quad (\alpha < \beta)$

$\therefore$ $\alpha^2 + \beta = 9 + 5 = 14$



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. Consider a parallel plate capacitor of area A (of each plate) and separation 'd' between the plates. If E is the electric field and ε_0 is the permittivity of free space between the plates, then potential energy stored in the capacitor is

(1) $\frac{1}{4}\varepsilon_0 E^2 Ad$ (2) $\frac{1}{2}\varepsilon_0 E^2 Ad$

(3) $\varepsilon_0 E^2 Ad$ (4) $\frac{3}{4}\varepsilon_0 E^2 Ad$

Answer (2)

Sol. We know energy density

$$\rho_{av} = \frac{1}{2}\varepsilon_0 E^2$$

So potential energy = $\rho_{av} \times$ Volume

$$\Rightarrow \text{P.E.} = \frac{1}{2}\varepsilon_0 E^2 Ad$$

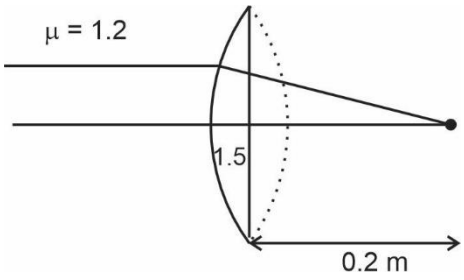
27. A thin plano convex lens made of glass of refractive index 1.5 is immersed in a liquid of refractive index 1.2. When the plane side of the lens is silver coated for complete reflection, the lens immersed in the liquid behaves like a concave mirror of focal length 0.2 m. The radius of curvature of the curved surface of the lens is

(1) 0.20 m (2) 0.15 m

(3) 0.25 m (4) 0.10 m

Answer (4)

Sol.



We have shown extrapolated figure for reflecting side as it went undeviated.

$$\frac{1.5}{v_1} - \frac{1.2}{\infty} = \frac{(1.5-1.2)}{R}$$

$$\frac{1.2}{0.2} - \frac{1.5}{v_1} = \frac{(1.2-1.5)}{-R}$$

$$\frac{1.2}{0.2} = \frac{0.3}{R} + \frac{0.3}{R}$$

$$\Rightarrow 6 = \frac{2\times 3}{10R} \Rightarrow R = \frac{1}{10}\text{m} = 0.10\text{ m}$$

28. For an experimental expression $y = \frac{32.3\times 1125}{27.4}$, where all the digits are significant. Then to report the value of y we should write

(1) $y = 1330$

(2) $y = 1326.2$

(3) $y = 1326.19$

(4) $y = 1326.186$

Answer (1)

Sol. $y = \frac{32.3\times 1125}{27.4} = 1326.18$

So we need to report to three significant digit.

So, y = 1330

